

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)  |  | $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$<br><br>award (1) for reactants and products<br>award (1) for balancing only if reactants and products correct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2               |     |     | 2     |       |      |
|          |     | (ii) |  | award (1) for any of following <ul style="list-style-type: none"> <li>limestone forms lime / quicklime</li> <li>calcium carbonate forms calcium oxide /</li> <li><math>\text{CaCO}_3</math> forms <math>\text{CaO}</math></li> <li><math>\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2</math></li> </ul> award (1) for any of following <ul style="list-style-type: none"> <li>lime / quicklime reacts with sand (to form slag)</li> <li>calcium oxide reacts with silicon dioxide to form slag</li> <li><math>\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3</math></li> </ul> award (1) for identification of one of the reaction types e.g. <ul style="list-style-type: none"> <li>thermal decomposition / breaks down with heat</li> <li>neutralisation</li> </ul> | 3               |     |     | 3     |       |      |
|          | (b) | (i)  |  | $\text{Fe}_2\text{O}_3(\text{s}) + 6\text{HCl}(\text{aq}) \rightarrow 2\text{FeCl}_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 | 1   |     | 1     | 1     |      |
|          |     | (ii) |  | $3\text{OH}^-(\text{aq}) + \text{Fe}^{3+}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$<br><br>award (1) for product<br>award (1) for balancing <b>only</b> if all formulae are correct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 | 2   |     | 2     |       |      |